

1. Cell Theory has THREE principles. Write each one in full and explain why it is important:

2. Robert Hooke first saw cells in 1665. Cell Theory was only formally established in the 1830s–1850s. Why do you think there was a gap of nearly 200 years? Write TWO sentences.

3. The lesson says THREE structures are found in plant cells but NOT in animal cells. For each one, complete the table:

4. Complete the full comparison table. Tick the correct cells and give the reason for any difference:

5. Use the formula: Total magnification = eyepiece lens × objective lens

a) Eyepiece = ×10, Objective = ×40. Total magnification = _____

b) A cell appears 3 mm on a slide at this magnification. What is its actual size in mm? Show working below:

Working: _____

Actual size = _____

c) Why should you ALWAYS start with the lowest objective lens ($\times 4$), never the highest?

6. A plant is wilting. Use the lesson to explain the chain of events. Use the word 'turgor pressure' in your answer. Write THREE sentences.

7. The lesson says mitochondria are 'the powerhouse of the cell.' Why do muscle cells have MORE mitochondria than skin cells? Write TWO sentences using the lesson's information.
